An Agent-Based Boom-Bust Business Cycle Model with Search-for-Yield and Heterogeneous Expectations in the Bond Market

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Abstract

This paper analyzes how heterogeneous expectations in the credit market amplify the transmission of shocks from the financial sector to the real economy. We adapt Chiarella and di Guilmi (2011) to include a credit market in which investors switch expectations according to the mechanism proposed by Brock and Hommes (1997). During boom phases more and more investors follow a trend-trading strategy. This search for yield lowers the risk premium, making firms take on more debt to finance more investment, which validates investors' expectations. Eventually, firms are over-leveraged, a small shock forces default and the economy enters the bust phase. This model offers a clear explanation for the behavior of risk premia over the business cycle, shows how they can be mis-priced and considers the welfare implications of the mis-pricing of risk. The model is first simulated using agent-based modelling techniques and then solved with stochastic aggregation methods.

Keywords: Business cycle; Fluctuations; Leverage; Default
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